Unit-I: INTELLIGENT AGENTS
Introduction to artificial intelligence; Intelligent agents: agents & environment, concept of rationality, nature of environments, structure of agents.
Case Study: Autonomous Delivery Robots which interact with their surroundings and navigate through dynamic environments to deliver packages.

Unit-II: PROBLEM SOLVING AGENTS
Uninformed search strategies, Heuristic search strategies, heuristic functions; Local search and optimization problems, local search in continuous space, search with nondeterministic actions, search in partially observable environments, online search agents and unknown environments.
Case Study: Autonomous vehicle Navigation in Unknown Environments

Unit-III: GAME PLAYING AND CSP
Adversarial search: Games, optimal decisions in games, alpha - beta pruning, stochastic games, partially observable games; Constraint satisfaction problems; constraint propagation, backtracking search for CSP, local search for CSP, structure of CSP
Case Study: Artificial intelligence system plays chess to make optimal moves in a partially observable and dynamic environment.

Unit-IV: LOGICAL AGENTS
Knowledge-based agents, propositional logic, propositional theorem proving, propositional model checking, agents based on propositional logic; First-order logic: syntax and semantics, knowledge representation and engineering; Inferences in first-order logic: forward chaining, backward chaining, resolution
Case Study: Automated personal assistant to assist users in managing their daily tasks, scheduling, and information retrieval.

Unit-V: KNOWLEDGE REPRESENTATION AND PLANNING
Ontological engineering, categories and objects, events, mental objects and modal logic, reasoning systems for categories, reasoning with default information; Classical planning, algorithms for classical planning; time, schedule, and resources analysis, hierarchical planning, planning and acting in non-deterministic domains
Case Study: Autonomous Warehouse Management System (WMS) for efficient planning, scheduling, and resource allocation within a warehouse environment.